

Mohammad Raditia Pradana

Depok, Indonesia
Researcher, Universitas Indonesia
M.Sc. in Geography Sciences

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RESEARCH INTERESTS

Street-view imagery (SVI) • Urban data science • Computer vision • Semantic segmentation • Geospatial analysis • Urban green spaces • Urban planning

EDUCATION

M.Sc. in Geography Sciences (Fast-Track Program) — Universitas Indonesia 2024 – 2025

GPA 4.00 / 4.00 | Summa Cum Laude, Faculty Level | Ranked 1st in Cohort

Thesis: Spatial Modeling of Visual Element Proportion Diversity in Urban Spaces Based on Street-View and Spatial Indicators (Kebayoran Baru & Tanah Abang)

B.Sc. in Geography (Fast-Track Program) — Universitas Indonesia 2020 – 2024

GPA 3.85 / 4.00 | Cum Laude, Department Level | Ranked 1st in Cohort

Thesis: Protection Level for Fire-Prone Areas in East Jakarta City

EXPERIENCE

Researcher May 2025 – Present

Department of Geography, Universitas Indonesia

- Co-PI on a university-funded grant (IDR 100 million) on climate resilience in urban parks via SVI and place attachment analysis.
- Mentors 3 undergraduate students across the full research cycle.

Full-time Researcher & Assistant Editor, Smart City Journal Oct. 2024 – Jun. 2026

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- Geospatial research using SVI and generative AI for urban environment analysis.
- Manages *Smart City Journal* editorial workflow (manuscript handling, reviewer coordination).

Cartographer — SEA, West Bangka Regency Mar. – Dec. 2025

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Solely responsible for producing 50+ thematic and base maps at multiple cartographic scales for the SEA document.

Part-time Research Assistant Jun. – Oct. 2024

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- GIS and remote sensing analysis using 3D data and digital building audits.
- Project-in-Charge for a government-collaborated urban research project.

Cartographer — Detailed Spatial Planning, Baradatu District May – Aug. 2024

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Compiled 20+ large-scale base and thematic maps at 1:5,000 for the RDTR planning document.

Cartographer — SEA, Bangka Belitung Islands Province Mar. – Oct. 2024

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Solely produced 40+ thematic maps including environmental sensitivity and spatial constraint layers.

Student Researcher Jun. 2023 – Oct. 2024

Department of Geography, Universitas Indonesia

- Fieldwork on urban fire management in Jakarta; produced 1 publication and 2 international conference presentations.

Cartographer — Regional Spatial Planning, Hulu Sungai Tengah Regency Jan. – Apr. 2023

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Produced 40+ thematic maps at 1:50,000 scale for the RTRW document, South Kalimantan.

TEACHING

Master Programme of Urban Planning, Universitas Indonesia

Guest Lecturer — Data Management and Analysis

Apr. – May 2026

- Delivered technical lectures on database development, statistical analysis, and spatial analysis for graduate urban planning students.

Department of Geography, Universitas Indonesia

Teaching Assistant — GIS & Remote Sensing Practicals

Jun. – Aug. 2024

- Facilitated hands-on practicals for 30+ undergraduates across 6 self-developed modules covering core GIS and remote sensing workflows.

PUBLICATIONS

Peer-Reviewed Journal Articles

- [1] **Pradana, M.R.** & Semedi, J.M. (2026). Indonesia's urban green spaces dilemma: Balancing law, ecology, and public space, highlighting Jakarta. *Human Geography*. [DOI]
- [2] **Pradana, M.R.**, Dimiyati, M. & Gamal, A. (2025). Harmonizing street-view semantics and spatial predictors for dominant urban visual composition modelling. *Computational Urban Science*, 5(5). [DOI]
- [3] **Pradana, M.R.**, Wibowo, A. & Semedi, J.M. (2025). Multi-perspective evaluation of urban green views: Spatial and street-view data integration in Sudirman Central Business District, Indonesia. *Geomatics and Environmental Engineering*, 19(3), 91–114. [DOI]
- [4] **Pradana, M.R.**, Dimiyati, M. & Semedi, J.M. (2025). Unveiling greenery and visual comfort: Integrating Green View Index and image segmentation in panoramic rural landscapes. *GeoScape*, 19(2), 143–159. [DOI]
- [5] **Pradana, M.R.** & Dimiyati, M. (2024). Tracking urban sprawl: A systematic review and bibliometric analysis of spatio-temporal patterns using remote sensing and GIS. *European Journal of Geography*, 15(3), 190–203. [DOI]
- [6] Gamal, A., **Pradana, M.R.**, Wibisono, B.H., Navitas, P. & Aryal, J. (2025). Thematic fragmentation and convergence in urban flood simulation research: A 45-year bibliometric mapping. *Urban Science*, 9(7), 280. [DOI]
- [7] Semedi, J.M., **Pradana, M.R.**, Putera, D.A. & Rahatiningtyas, N.S. (2024). The impedance of East Jakarta road network to estimate fire station service areas. *Forum Geografi*, 38(2), 280–290. [DOI]

Conference Proceedings

- [1] **Pradana, M.R.** & Semedi, J.M. (2025). Effortless coastal monitoring: Unsupervised detection of shoreline alterations due to tin mining in Bangka Belitung. *IOP Conference Series: EES*, 1462(1). [DOI]
- [2] Ash-Shidiq, I.P., Supriatna, S., Darmawan, A., Warta, Z., Molidena, E., Valla, A., Hisan, Firdaus, M.I., Zakaria, N.A., Muafiroh, S., **Pradana, M.R.**, Putera, D.A. (2024). Biomass stock estimation using Landsat 8 imagery in Bukit Tiga-puluh National Park, Riau. *IOP Conference Series: EES*, 1291(1). [DOI]

Under Review

- In Review** Your Academic Career Depends on Your Co-Authors: When Single-Author Papers Do Not Count in Indonesia's Academic Promotion System. **Journal of Academic Ethics**.
- In Review** Reframing Urban "Sustainable" Streetscapes: Evidence from Cross-City Transformation and Convergence. **City and Environment Interactions**.
- In Review** Interpreting 'Sustainable' Urban Streetscapes with Generative AI: Context-Rich vs. Generic Prompting. **Urban Informatics**.

RESEARCH & GOVERNMENT PROJECTS

Research Grants

Prompting Sustainability: A Comparative Analysis of How Generative AI Interprets and Visualizes Urban 'Sustainable' Streetscapes Oct. 2025 – Present

Funded by Universitas Indonesia

Cooling Cities, Warming Communities: Quantifying Climate Resilience in Urban Parks Through Anthropomorphic Mapping and Place Attachment May 2025 – Present

IDR 100 million, 2-year grant — Universitas Indonesia

Spatial Modeling of Visual Element Proportion Diversity [Master's Thesis]

Jun. 2024 – Jun. 2025

INSPIRASI Grant — SMART CITY UI

Collaborative Mapping & Mitigation for a Resilient Capital City with Generative AI PRIME Engineering Grant (seed; potential IDR 1B follow-up)	Jun. 2024 – Dec. 2025
Level of Protection of Fire-Prone Areas in East Jakarta City Funded by Faculty of Mathematics and Natural Sciences, Universitas Indonesia	Aug. 2023 – Jan. 2024

Government Projects

Innovative Government Awards 2025, Ministry of Home Affairs (<i>Lead data scientist; 12,000+ data assessed</i>)	Sep. – Nov. 2025
Strategic Environmental Assessment (SEA) — West Bangka Regency	Mar. – Dec. 2025
Innovative Government Awards 2024, Ministry of Home Affairs (<i>Lead data scientist; 8,000+ data assessed</i>)	Sep. – Oct. 2024
Strategic Environmental Assessment (SEA) — Bangka Belitung Islands Province	Mar. – Oct. 2024

OPEN-SOURCE SOFTWARE & DATASETS

Software & Tools

Cooling Cities — Research dissemination website	2025 
Prompting Sustainability — Research dissemination website	2025 
GSV Panorama Downloader — Notebook repo for batch downloading Google Street View panoramas	2025 
GSV Downloader by Links — Notebook repo for downloading Street View imagery via direct links	2025 

Open-Access Datasets

genai_sustainablestreetscape (v2) — Cross-city corpus of 4,186 paired images (raw + GPT-5-generated counterparts) from Jakarta (567), Melbourne (832), and Singapore (694). Standardised to 640×640 px; includes source coordinates and generation metadata. Each paired with semantic segmentation masks.	2026 
Sustainable Streetscape Prompt-Specificity Dataset (v1) — 100 Jakarta GSV scenes × 3 treatments (raw, context-rich, non-context via GPT-4o), each paired with semantic segmentation masks.	2026 

PEER REVIEW SERVICE

<i>Discover Environment</i> , Springer Nature	June 2026
<i>Journal of Mountain Science</i> , Springer Nature	May - June 2026
<i>Scientific Reports</i> , Nature Publishing Group	Jan. – June. 2026
<i>Frontiers in Marine Sciences</i> , Frontiers	Jan. – Feb. 2026
<i>Urban Forestry & Urban Greening</i> , Elsevier	May – Aug. 2025
<i>Transactions in GIS</i> , Wiley	Jan. 2025

CONFERENCE PRESENTATIONS

Aug. 2024	ICoAGPG — <i>Effortless Shoreline Detection in Tin-Mining Areas</i> . [Best Presenter Award]
Nov. 2023	AICGEE — <i>The Impedance of East Jakarta Road Network to Estimate Fire Station Service Areas</i> .
Nov. 2022	ICoSAG — <i>Biomass Stock Estimation Using Landsat 8 Imagery in Bukit Tigapuluh National Park</i> .

HONORS & AWARDS

2025	Ranked 1st in Cohort, M.Sc. Geography — Summa Cum Laude (Faculty Level)
2024	Ranked 1st in Cohort, B.Sc. Geography — Cum Laude (Department Level)
2024	Fast-Track Program Candidacy, B.Sc. - M.Sc. Geography — selected based on academic excellence during undergraduate study
2024	Best Presenter Award, ICoAGPG 2024
2022	2nd Place, National Economics Paper Competition (IMIE FEB UKSW) — spatial-economic policy aligned with SDGs

Urban Analytics	Street-view imagery analytics, semantic segmentation, urban environmental assessment, geospatial data science, spatial pattern analysis
Programming	Python (familiar), R (familiar)
Geospatial Platforms	ArcGIS Pro, QGIS, Google Earth Engine
Research Computing	Google Colab, Jupyter Notebook, VS Code, Git, GitHub, $\LaTeX$
Quantitative Methods	Statistical analysis, machine learning, computer vision, dimensionality reduction, systematic literature review, bibliometric analysis